Answer Key & Teacher Notes

Session 1 — Building a Box Plot · Session 2 — Comparing Box Plots

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Building a Box Plot

TALK PROMPTS

- What part of the data does the box of a box plot cover?
- What does a long box tell you about the middle half of the counts?

PARTNER TASK

Bird Counts. One partner orders the seven counts and finds the median; the other finds Q1 and Q3 from the halves. Together draw the box plot and check that each section holds about one fourth of the values.

WATCH FOR

- Using the values next to the median as Q1 and Q3.
- Including the median in both halves when the count is odd.
- Thinking a long whisker or section holds more of the data than a short one.

SESSION 2 · TEACHER NOTES

Comparing Box Plots

TALK PROMPTS

- Why do two box plots let you compare groups without listing every value?
- Which bus route's times are more alike, and how does the box show it?

PARTNER TASK

Two Teams. Each partner finds the five-number summary for one team in problem 2. Draw both plots on one scale, then compare medians and box widths together.

WATCH FOR

- Comparing whisker lengths instead of median lines to judge center.
- Thinking the same box width means the same spread when the whiskers differ a lot.
- Comparing two box plots drawn on different scales.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.4	1	1	2	2	1	1	2	3	2	1	1
S1 · Form B	6.SP.B.4	1	1	2	2	1	1	2	3	2	1	1
S2 · Form A	6.SP.B.4	1	2	2	2	3	1	3	3	2	1	1
S2 · Form B	6.SP.B.4	1	2	2	2	3	1	3	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

five-number summary: Min, Q1, **median**, Q3 and max.

quartile: Q1 is the median of the **lower** half; Q3 is the median of the upper half.

box plot: A graph of the five-number summary: a box from Q1 to Q3 and two **whiskers**.

- With an odd number of values, leave the **median** out of both halves.
- Each of the four sections of a box plot holds about one **fourth** of the data.

Example: Order: 5, 9, 11, 14, 16, 20, 23. Median = **14**. Lower half 5, 9, 11: Q1 = **9**. Upper half 16, 20, 23: Q3 = **20**. Answer: 5, 9, 14, 20, **23**

YOU TRY

1. Order: 3, 5, 7, 8, 10, 12, 15. a. 5 b. 8 c. 12

2. 10; 16; 22

EXIT TICKET

1. Order: 2, 4, 6, 7, 9, 11, 13. a. 4 b. 7 c. 11

2. 14 − 6 = 8

3. **B** (1/2)

PRACTICE FORM A

1. Order: 12, 15, 18, 21, 24, 26, 30. a. 12 b. 15 c. 21 d. 26 e. 30 Box from 15 to 26, line at 21, whiskers to 12 and 30.

2. a. 9 b. 11 c. 15 d. 20 e. 25 f. 20 − 11 = 9

3. No. Each of the four sections holds about one fourth of the data, so 1/4 is between 20 and 27, not half.

4. a. 34 − 23 = 11 b. about half (1/2)

5. Order: 18, 25, 29, 33, 40, 47, 52. a. 18 b. 25 c. 33 d. 47 e. 52

6. a. 18 b. 40. Box from 18 to 31 with a line at 24; whiskers to 12 and 40.

7. Order: 44, 48, 50, 52, 57, 59, 61, 65. Halves: 44, 48, 50, 52 | 57, 59, 61, 65. a. 44 b. 49 c. 54.5 d. 60 e. 65

8. a. Order: 12, 15, 18, 20, 22, 25, 27, 30, 33. Min 12, Q1 16.5, median 22, Q3 28.5, max 33 b. The middle half of the costs is from $16.50 to $28.50. c. A long box means the middle costs are spread out; a short box means they are close together.

9. Yusuf used the values next to the median. Q1 is the median of the lower half (5, 9, 12), so Q1 = 9. Q3 is the median of the upper half (20, 24, 30), so Q3 = 24.

10. **B** (Q1 to Q3)

11. **C** (6)

Self-check word **QUARTILE**

PRACTICE FORM B

1. Order: 12, 14, 16, 19, 23, 27, 30. a. 12 b. 14 c. 19 d. 27 e. 30 Box from 14 to 27, line at 19, whiskers to 12 and 30.

2. a. 8 b. 10 c. 16 d. 21 e. 24 f. 21 − 10 = 11

3. No. Each of the four sections holds about one fourth of the data, so 1/4 is between 31 and 44, not half.

4. a. 35 − 20 = 15 b. about half (1/2)

5. Order: 15, 22, 27, 30, 36, 44, 49. a. 15 b. 22 c. 30 d. 44 e. 49

6. a. 29 b. 10. Box from 16 to 29 with a line at 23; whiskers to 10 and 38.

7. Order: 42, 46, 49, 53, 55, 58, 60, 67. Halves: 42, 46, 49, 53 | 55, 58, 60, 67. a. 42 b. 47.5 c. 54 d. 59 e. 67

8. a. Order: 11, 14, 16, 21, 24, 27, 29, 32, 35. Min 11, Q1 15, median 24, Q3 30.5, max 35 b. The middle half of the costs is from $15 to $30.50. c. A long box means the middle costs are spread out; a short box means they are close together.

9. Marisol used the values next to the median. Q1 is the median of the lower half (4, 7, 11), so Q1 = 7. Q3 is the median of the upper half (18, 23, 28), so Q3 = 23.

10. **B** (lower half)

11. **C** (25)

Self-check word **BOXPLOTS**

GUIDED NOTES

center: A typical value; on a box plot, the **median** line.

box width: Q3 **minus** Q1; the spread of the middle half.

consistent: Values are **close** together; the box is narrow.

- Compare **medians** to see which group has the higher typical value.
- Compare box widths: the **wider** box is more spread out.

Example: Medians: 26 − 20 = **6**, so Y is higher by 6 min. Box widths: X is **4**, Y is **14**. Answer: Trail **X** is more consistent.

YOU TRY

1. B has the higher median (11 vs. 9); A is more spread out (box 6 vs. 5)

2. a. 18 b. 6

EXIT TICKET

1. a. 12 b. 6 c. K

2. Class B, by 7

3. **B** (S is more spread out)

PRACTICE FORM A

1. a. Quiz 2 b. 10 − 9 = 1 c. Quiz 2 (7 vs. 4)

2. Red: 42, 46, 50, 54, 56 (box 8). Blue: 37, 41, 45, 55, 58 (box 14). More consistent: Red

3. a. 6 b. 24 (D is more spread out)

4. P. Its box is 8 wide; Q's box is 18 wide.

5. No. M's whole span is 90 − 20 = 70 and its box is 65 − 45 = 20. N's span is 70 − 50 = 20 and its box is 60 − 55 = 5. M is much more spread out.

6. Group B, by 57 − 53 = 4

7. a. 60 − 10 = 50 b. 44 − 28 = 16 c. Only partly: the middle halves match, but W's whiskers stretch much farther, so W is more spread out overall.

8. a. Route 2 (29 vs. 28 min) b. Route 2 (box 5 min vs. 9 min) c. Route 2: its times are close together, so a rider can count on it.

9. The whisker shows how far the greatest values reach, not the center. To compare centers, compare the median lines inside the boxes.

10. **B** (median)

11. **A** (12)

Self-check word **SPREADS**

PRACTICE FORM B

1. a. Quiz 2 b. 11 − 8 = 3 c. Quiz 2 (9 vs. 6)

2. Gold: 35, 38, 43, 47, 49 (box 9). Green: 30, 33, 40, 50, 52 (box 17). More consistent: Gold

3. a. 5 b. 22 (D is more spread out)

4. P. Its box is 9 wide; Q's box is 17 wide.

5. No. M's whole span is 80 − 10 = 70 and its box is 55 − 40 = 15. N's span is 60 − 35 = 25 and its box is 50 − 44 = 6. M is much more spread out.

6. Group B, by 38 − 31 = 7

7. a. 65 − 5 = 60 b. 41 − 22 = 19 c. Only partly: the middle halves match, but W's whiskers stretch much farther, so W is more spread out overall.

8. a. Route 1 (29 vs. 28 min) b. Route 1 (box 4 min vs. 14 min) c. Route 1: its times are close together, so a rider can count on it.

9. The whisker shows how far the greatest values reach, not the center. To compare centers, compare the median lines inside the boxes.

10. **A** (more spread out)

11. **C** (25)

Self-check word **COMPARE**

SHEET 1 · BUILDING A BOX PLOT

1. Ordered: 10, 14, 18, 22, 25, 30, 34, 40, 46. Min 10, Q1 16, Median 25, Q3 37, Max 46.
2. Whiskers from 10 to 16 and 37 to 46; box from 16 to 37; median line at 25.
3. a. 64°F b. Q1 = 58°F, Q3 = 70°F c. About 1/2 of the days (the middle half)
4. Ordered: 11, 14, 18, 20, 26, 28, 32, 39 (even count: halves are 11, 14, 18, 20 and 26, 28, 32, 39). a. Q1 = (14 + 18) ÷ 2 = 16 b. Median = (20 + 26) ÷ 2 = 23 c. Q3 = (28 + 32) ÷ 2 = 30
5. With 7 values the median (8) is left out of both halves. The lower half is 2, 5, 7, so Q1 = 5.

SHEET 2 · COMPARING BOX PLOTS

1. a. Class B, by 5 min (30 vs. 25) b. A: 30 − 15 = 15 min; B: 45 − 20 = 25 min c. Class A (narrower box)
2. a. 30 − 12 = 18 b. 41 − 5 = 36
3. Maya: box width 110 − 95 = 15, range 125 − 80 = 45. Leon: box width 112 − 90 = 22, range 140 − 70 = 70. Maya is more consistent (smaller box width and range).
4. Group T. Its box is 20 − 8 = 12 cm wide; Group S's box is 16 − 12 = 4 cm wide. A wider box means more variation.
5. Bus A. Both have a typical time of 20 min, but Bus A's box is 22 − 18 = 4 min wide and Bus B's is 26 − 14 = 12 min wide. Bus A's times are more consistent, so it is easier to plan for.
6. **B** (Y)
7. **A** (9)

October 2026 conceptual additions — teacher key

Session 1 · Building a Box Plot

Minimum 2, Q1 4, median 8, Q3 12, maximum 14. The box extends 4 to 12, with a median line at 8 and whiskers to 2 and 14.

Extension: A box plot records five summaries, not every observation. Different data sets can share those summaries.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Comparing Box Plots

A has median 8 and IQR 4; B has median 10 and IQR 8. A has the lower median and smaller middle-half spread. Sample sizes cannot be determined from these box plots alone.

Extension: Equal box lengths show equal IQR only; medians, extremes, individual observations, and counts could differ.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.